part. Vessels are developed in this embryonic zone which can only be recognised by the presence of giant cells and by its general disposition. The caseous material is, as it were, encysted. It may undergo ulterior changes, which, by the infiltration of calcareous particles, convert it into a sort of putty, and, in the most advanced degree, into a small calculus. It is no longer possible for this caseous centre to be resorbed and disappear. This encysting process is marked by a certain amount of subacute inflammatory irritation, which is communicated to the neighbouring parts; also, around this healing tubercle a whole series of alveoli occur, marked off from the rest by their thickened fibrous walls, whilst their interior is lined by cubical epithelium, just as in cases of primary broncho-pneumonic cirrhosis.

The same applies to the peribronchial tuberculous agglomerations of phthisis with bulky products. If the curative process is but slightly marked, the two stages of fibrous evolution just mentioned may be followed in the ordinarily visible embryonic zone, ending in the encysting of caseous products precisely as in the case of miliary tubercle. The following is a disposition often met with in the lungs of subjects who, after having presented signs of pulmonary phthisis, have returned to health and succumbed to some intercurrent disease. It presents the appearance of tuberculous agglomeration in full development. The caseous zone is distinct, the elastic fibres still visible; but the embryonic zone is reduced to a fibrous cystic wall. The presence of giant-cells, which persist, places the interpretation given by M. Charcot almost beyond dispute.

The mechanism of the formation of healing cavities has been discussed above.

In lymphatic glands, in the laryngeal mucous membrane, in the skin of the face (so-called scrofulous lupus), *stationary* tubercles are frequently seen. Here caseous mortification is not necessarily a fatal affair. The tubercle, arrived at its full development, whether this be an elementary or an agglomerated tubercle, persists as such, and, like a parasite, preserves the integrity of its structure. Tubercles of this kind of evolution have in general a complex and very regular structure; giant-cells, epitheliod cells, the reticulum (which caused a belief in the existence of a vestige of lymphatic organisation), are very marked.

In the *fibrous tubercle* there is no longer a partial fibrous evolution taking place in the embryonic zone, around the caseous zone, as in ordinary tubercle. But there is tubercle presenting this peculiarity, that it very early loses its cellular character, and undergoes a special evolution, transforming it rapidly into a small fibrous tumour, without there ever being the slightest trace of caseous mortification. Fibrous tubercle is not then the adult grey granulation; but it does not differ from it integrally, and in that Laennec was right. Fibrous tubercle exists under two chief forms—miliary tubercle and tuberculous agglomeration. The fibrous miliary tubercle is thus characterised anatomically. In a lung otherwise healthy, presenting at its apex the characters of fibroid phthisis, are found granulations presenting all the characters so clearly pictured in Bayle's description of grey granulations; these granulations are grouped in the form of bunches of grapes, of which the bronchi seem to represent the stalk. The nodules composing the grapes have the following composition:—Agglomeration of numerous primary tubercles, concentric disposition of zones, giant-cells often recognisable, an absolutely fibrous structure, nowhere caseous. Other nodules in the same preparation still present the embryonic structure, always with marked predominance of the intercellular fibrous substratum, without a trace of central mortification. The lung-tissue in the vicinity presents, besides, thickening of the interlobular spaces and interlobular bands; in a word, tubercle and concomitant pneumonia, everything pointing to a primitive chronic process.

There is also found, as has been said, in phthisis essentially chronic, fibrous tubercle under the form and with the characters of tuberculous agglomerations. The nodules are more voluminous. There is a peripheral fibrous zone, and a central zone equally fibrous, where, however, are to be seen traces of the artery, bronchus, and alveolar cells. But it is all fibrous—no part is caseous. M. Charcot remarks that it is very difficult to admit that the central zone, after having undergone caseous metamorphosis, has been—after the elimination and resorption of the caseous products—the seat of a reparative fibrous evolution. Caseous products cannot organise. It is most likely that the fibrous evolution takes place when the whole tubercle is in the condition it ordinarily presents: embryonic zone; alveolar walls infiltrated with cells; alveoli filled with polypiform embryonic products. If we suppose this embryonic mass to wholly undergo fibrous evolution, we shall have the representation of that which the fibrous agglomerate tubercle exhibits on microscopical examination. Where fibrous tubercle preponderates, and *à fortiori* where it exists nearly exclusively, there we have *fibroid phthisis*. The lesions of this anatomical variety of phthisis may be thus summed up: lung diminished in volume, presenting over a variable extent in its upper lobe the characters of pulmonary cirrhosis; here and there a few cavities, a few healing tubercles, but especially fibrous tubercles. The phrase "fibroid phthisis" should be reserved for cases where the induration extends through the whole, or nearly the whole, of one lobe. The process of fibrous transformation is often met with limited to a few parts of the parenchyma in all the forms of phthisis of rather slow evolution; according to Thaon, it would occur in one-third of the cases. This fibroid phthisis, to which certain tuberculous subjects seem more particularly predisposed, is not solely characterised, from the clinical point of view, by phenomena of dilatation of bronchi, but by the displacement of neighbouring organs, which may be also explained by changes that supervene in the physical constitution of the organ.

(*To be concluded.*)

MEDICAL CHARGES.

AT the Chelmsford County Court, on Dec. 16th, a case of some importance to medical men came before the judge, Mr. T. J. Abdy, D.C.L. The plaintiff, Dr. Nicholls, of Chelmsford, sought to recover from a merchant of London an account for medical attendance upon his wife and for medicine supplied.—Mr. T. F. Woodward, solicitor, appeared for the plaintiff, and Mr. Canday, barrister-at-law, for the defendant.—It appeared from the statements made that the defendant and his wife separated by mutual consent in February, 1879, an allowance of £13 per month being made to the lady. Later in the year she took up her residence with a lady friend at Chelmsford, and while there, in the month of October, she was seized with a serious illness. Dr. Nicholls was sent for, and he found her almost at the point of death. He continued to attend her till March, 1879, and she recovered.—Mr. Canday contended that as the wife had a reasonable allowance, which had been regularly paid, she had no power to pledge her husband's credit, and that, therefore his client was not liable for the amount sought to be recovered.—His Honour said he had only one case to guide him, and that was in favour of the husband. Professional men and tradesmen, before trusting a woman living apart from her husband, should ascertain whether they would be likely to recover from the husband. He must give his verdict for the husband, but against the wife (who had been made a party in the case).—Mr. Canday: Now that my client's non-liability has been established, he will give Dr. Nicholls a cheque for the amount.—His Honour: That is most generous.—The costs of the defendant and his advocate were allowed, and the Judge ordered the wife to pay Dr. Nicholls' costs.

Correspondence.

"Audi alteram partem."

NERVE-STRETCHING IN TETANUS.

To the Editor of THE LANCET.

SIR,—My attention has been pointedly drawn to a letter on this subject in your issue of the 13th inst., from an American surgeon, criticising a clinical lecture I delivered on a case in which I stretched the great sciatic nerve for tetanus following contusions of the foot.

I would wish your readers who are interested in this matter to do me the justice to form their own conclusions from my paper as to the basis on which I founded my opinion, rather than from the comments made upon it in Mr. Johnstone's letter.

At page 933 of the *British Medical Journal* for June 21, 1879, will be found an account of the case, with *a very brief abstract* of my lecture upon it. It is there stated that, after the stretching, "the course of the symptoms seemed more severe and more rapid. Although Mr. Morris refused, with the present slight data upon which to form an opinion, to pledge himself to any decided conclusion, he could not help expressing his conviction that nerve-stretching in tetanus would not prove a favourable mode of treatment. It seemed to him unscientific in principle, and, so far, most unsatisfactory in practice. The idea was to destroy the conductivity of the trunk-nerve, so as to cut the spinal cord off from its communication with the peripheral nerves at the seat of injury. But, in doing so, a very severe shock, or irritation, was likely to be excited by the nerve-stretching itself; so that the condition of the cord, be it what it might—whether a material change, a state of excitement from irritation, or a 'bad habit' induced by a 'depraved current'—seemed to him likely to be aggravated, not mitigated, by a new and great disturbance of the same nerve-channel, at a spot much nearer to the spinal centre. But at present the matter must be considered as *sub judice*."

Had the lecture been published in full it would have been found that I pointed out at the time that in order to have completely carried out the treatment by nerve-stretching the anterior crural, or at least the long saphenous nerve ought to have been pulled as well as the sciatic; and that, as this was not done, incisions through skin and subcutaneous tissue were made, encircling the inflamed wounds. The sciatic was the chief source of the nerve supply to the part, and therefore was selected; and though it requires but a poor knowledge of anatomy to know that the long saphenous nerve supplies the skin on the inner side of the ankle and foot, it would have required a much greater faith than I possess in the benefits likely to result from nerve-stretching in tetanus to have turned the patient over and cut down upon his anterior crural immediately after I had witnessed "a very severe general spasm" follow directly upon the pulling of the great sciatic.

That I acted thoroughly under the impression of the necessity of operating early and of procuring "complete isolation" from the wound, must be admitted from the fact that immediately I saw my patient I suggested amputation. As the mother refused I at once decided on trying the effect of nerve-stretching. The operation was performed within eighteen hours of the first spasm, but the child was dead within twenty-five hours of the onset of the attacks; and, as I have stated, the progress of the case appeared to have been hastened after the operation.

It was and is far from my wish "to prejudice scientific men against the only proceeding that promises much in the management of tetanus," and I shall be very glad to have reason to change my conviction. But as the number of operations reported is still small, and as, besides the ten unsuccessful cases to which your correspondent refers, death has also followed in two cases operated upon by Mr. Nankivell, in one by Mr. Langton, in one by Mr. Hutchinson, in one out of two cases by Verneuil, and in one by Kocher; and, further, as in the two successful cases, of the details of which I am aware—viz., Verneuil's and Clark's—the symptoms were limited and the disease of a chronic nature; and, finally, as in the successful case of Ratton, the patient was kept under the influence of chloral for a fortnight, I feel bound to say that I do not yet see grounds for changing this conviction. Success will doubtless sometimes follow most modes of treatment if the disease be chronic, and the patient seen early. I have myself had one happy result with opium lotion applied to the injured part, and cold to the spinal column. In Mr. H. E. Clark's case other means of treatment, especially Calabar bean in large and frequent doses, were also employed, and he states that he is "inclined to take a very moderate view of the part taken by the operation in the cure of the patient," and adds that an attack followed the operation immediately. To me it appears very improbable that nerve-stretching can be so successful in the treatment of tetanus as nerve section, such, for example, as occurs in amputation; and though success has on some occasions followed amputation, most surgeons are fully conscious of the slight hope it holds out in cases of acute tetanus.

In conclusion, I think, Sir, it would have been well if your correspondent, writing in the interests of scientific accuracy, had stated more precisely the position of the burn in the unsuccessful case in his own practice. By simply saying the injury was "below and in front of the knee," he leaves it open to his readers to question how he could have secured *complete* isolation by stretching the trunks of only two out of three nerves which supply the parts about and below the knee.

I remain, Sir, your obedient servant,

HENRY MORRIS,
Surgeon to, and Lecturer on Anatomy at, the Middlesex Hospital.

Mansfield-street, Dec. 16th, 1879.

VACCINATION AND SMALL-POX AT GRAVESEND.

To the Editor of THE LANCET.

SIR,—Whilst the discussion on the subject of Animal Vaccination — or, as it would be more appropriate and classical to term the operation, Vituline Inoculation, if the latter word had not been so identified with small-pox—is postponed, it may not be uninteresting to give a short account of what has occurred in this town (Gravesend) during the late epidemic of small-pox. The disease was widely spread. Cases occurred in all parts of the town, but principally in the poorer.

The number of inhabitants is now about 24,000. The number of cases was not quite 200, the deaths 29. Mortality almost invariably occurred amongst those who came under the following classes—(1) the unvaccinated; (2) the insufficiently vaccinated; (3) those not revaccinated; (4) the dissipated. And I would also add that the very timid always appeared more liable to the disease, though I would not say that mortality was more common from this cause. Small-pox occurred mostly, though not entirely, amongst the poorer classes (which circumstance speaks for itself), and those living in close and unhealthy parts of the town.

Primary vaccination is nearly universal, and during the alarm thousands availed themselves of revaccination, with what effect in both the number of cases and deaths show plainly. Judging from experience obtained years ago, this number, if multiplied by ten, making 2000 cases and 290 deaths, would have been much nearer the mark if such had not been the case.

The table appended shows the ages of both recoveries and deaths :—

Table of Cases of Small-Pox in Epidemic of 1878.

Age.	Cases.	Death.
Under 1 year	4	4*
1 to 5	2	2†
5 to 10	30‡	0
10 to 14	16	3
14 to 20	50	5
20 to 30	40	6
30 to 40	38	4
40 to 50	16	4
50 to 60	2	1
Total	198	29

Vaccination here is commonly practised from arm to arm, and it is not often that any other than human lymph is used. I myself, in the course of thirty-five years' tenure of office as public vaccinator, have three or four times resorted to vituline lymph, procured from Dr. Warlomont, of Brussels, or Mr. Green, of Birmingham, but with no better success than with my old lymph. Indeed I have found it necessary to discontinue the use of such lymph from the excessive irritation it produced after a few transmissions (as was stated at the Medical Society), and which appeared to me to be unnecessary. The lymph which I have occasionally procured from the National Vaccine Establishment has always proved itself to be of the very best character. It has nearly always produced all the result I could wish for; and my only cause for using the word "nearly" is that once it failed me altogether when there was a very hard run upon the office on account of the great general anxiety for revaccination. None that I have ever received from this source has produced any unhealthy results. One gentleman at the Medical Society seemed to think that a supply should be forthcoming from the National Establishment whenever

* All unvaccinated.
† One unvaccinated. The other died of convulsions, and after death three or four pustules were found.
‡ All vaccinated.